PAYMENT MANAGEMENT SYSTEM

Project Documentation – Explanation Only

1. Introduction

The Payment Management System is a part of the E-Commerce Order Management System. It stores and manages payment details connected to customer orders. It helps track payment methods, transaction amounts, dates and payment statuses.

2. Payment Table Design

The Payment table contains the following fields:

Field	Explanation
Payment_ID	A unique number used to identify each payment record.
Order_ID	Identifies the order related to the payment.
Payment_Date	The date on which the payment was recorded or processed.
Payment_Mode	The method used to pay, such as UPI, Credit Card, Debit Card, Net Banking or Cash on Delivery.
Payment_Status	Shows whether the payment is Pending, Success or Failed.
Transaction_Amount	The amount associated with the payment transaction.

3. Relationships and Constraints

The Payment table is connected to the Orders table through Order_ID. This relationship ensures that payment information belongs to an order in the existing system.

Primary Key: Payment_ID uniquely identifies each payment record.

Foreign Key: Order_ID links a payment to an existing order.

NOT NULL: Ensures that required fields are not left empty.

UNIQUE: Ensures that an Order_ID appears only once in the Payment table. Therefore, each order can have at most one payment record.

CHECK: Restricts payment modes and statuses to the permitted values and prevents negative transaction amounts.

4. Order–Payment Relationship (ER Diagram Explanation)

One order can have zero or one payment record. It may have no payment record if payment details have not yet been entered. A payment record must be linked to an existing order. The UNIQUE constraint on Order_ID prevents multiple payment records for the same order.

ORDERS (Order_ID) 1 ———— 0..1 PAYMENT (Payment_ID, Order_ID)

5. Managing Payment Transactions

Adding payment records: Payment details are stored for orders using the order ID, date, payment mode, status and transaction amount.

Payment modes: The system supports UPI, Credit Card, Debit Card, Net Banking and Cash on Delivery.

Payment status updates: A payment can be marked Pending while awaiting confirmation, Success after confirmation, or Failed when the transaction does not complete.

Retrieving payment details: Payment information can be viewed for a particular order using its Order_ID.

Maintaining the relationship: Linking Payment to Orders keeps payment records associated with the existing Order Management System.

6. Payment Transaction Operations

Insert new payment transactions: Add a payment record for an existing order that does not already have a payment record.

Display successful payments: Show all transactions whose status is Success.

Find failed and pending transactions: Identify transactions that need review, confirmation or another payment attempt.

Update status after retry: After a retry, update the payment status according to the transaction result.

Count successful and failed payments: Calculate how many payments completed successfully and how many failed.

Display complete payment information: Combine payment, order and customer information to show the full transaction context.

7. Payment Analysis Reports

Transactions by payment mode: Counts how many transactions use each payment method.

Most preferred payment method: Identifies the method used for the greatest number of transactions. If methods are tied, all tied methods can be reported.

Total successful payment revenue: Adds the transaction amounts for successful payments only.

Revenue by payment method: Compares successful payment revenue across UPI, cards, Net Banking and Cash on Delivery.

Average transaction amount: Calculates the average amount across recorded payment transactions.

Customer payment history: Shows the orders and payment details associated with each customer.

Payment status summary: Counts successful, failed and pending transactions.

Orders with failed or pending payments: Lists orders whose payment status requires follow-up.

8. Sample Data Description

Sample records should include different payment modes and statuses. For example, one order may use UPI with a successful status, another may use a Credit Card with a successful status, and other orders may have failed or pending payments. The sample order IDs must match records already present in the Orders table.

9. Conclusion

The Payment Management System helps organize payment records, monitor transaction status and prepare useful revenue reports. The relationship with the Orders table maintains consistency across the E-Commerce system, while the constraints prevent duplicate payment records for the same order and restrict invalid payment values.